Young modulus room temprature

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1 Introduction

The objective of this experiment is to determine Young's modulus E by finding the natural frequency of a cantilever beam, which is made of Carbon Fiber Reinforced Polymer (CFRP). To verify our method, we utilized four samples made with four different materials, Aluminum, Stainless steel, Copper, and Brass. In parallel, we use CFRP, which is our target as the fifth sample.

Figure 1 shows the setup of the experiment. It can be seen that the beam is fixed by two screws at one end and is free to vibrate at the free end. In addition, the whole setup is fixed to the table by two metal clamps. A guitar pickup is used as a coil to generate the magnetic field around the beam. As the beam starts to vibrate (excited by the hammer), according to Faraday's law of induction, Equation (44), a changing magnetic flux induces a voltage, and the magnitude of that voltage is proportional to the rate of change of the flux.

$$\mathcal{E} = -N\frac{d\Phi_B}{dt}, \tag{1}$$

Where $\mathcal{E}$ is the induced electromotive force (emf), N is the number of turns in the coil, Φ_B is the magnetic flux through one loop, and t is time. The minus sign indicates the direction of the induced emf according to Lenz's law. These changes are recorded by the oscilloscope as a voltage-time signal. The Fast Fourier method is used to convert this signal into the frequency domain. After finding the natural frequency, the Young modulus can be found. We use two different scenarios to measure the Young modulus. In the first scenario, we load the free end by hanging masses. In the second scenario, we leave the free end without mass. For a cantilever beam with a tip mass m_{tip}, the fundamental frequency is given by

$$f = \frac{1}{2\pi}\sqrt{\frac{3EI}{(m_{\text{tip}} + 0.235.m_{\text{beam}})L^3}}, \tag{2}$$

which can be rearranged into a linear form:

$$\frac{1}{f^2} = \frac{4\pi^2 L^3}{3EI}\left(m_{\text{tip}} + 0.235.m_{\text{beam}}\right). \tag{3}$$

Thus, plotting $1/f^2$ versus m_{tip} yields a straight line, where the slope is inversely proportional to E. For a cantilever beam without a tip mass, the fundamental frequency is

$$f_0 = \frac{1.875^2}{2\pi L^2}\sqrt{\frac{EI}{\rho A}}, \tag{4}$$

which leads to

$$f_0^2 = \left(\frac{1.875^4}{4\pi^2}\right)\frac{EI}{\rho A}\frac{1}{L^4}. \tag{5}$$

Therefore, plotting f_0^2 versus $1/L^4$ yields a linear relation, from which E can be extracted.

By comparing both approaches, we assess the consistency of the results and the agreement with theoretical expectations. Measurements were repeated for multiple tip masses and three different beam lengths. Each configuration was measured multiple times to estimate statistical uncertainty.

For each beam length, the quantity $1/f^2$ was plotted against the tip mass. A weighted linear fit was performed to extract the slope. Young's modulus was computed using the relation

$$E = \frac{4\pi^2 L^3}{3I \cdot \text{slope}}. \tag{6}$$

Uncertainties were propagated using a Monte Carlo method, taking into account measurement errors in length, geometry, and fitted parameters. The squared fundamental frequency f_0^2 was plotted as a function of $1/L^4$. A linear fit constrained through the origin was applied. Young's modulus was calculated using

$$E = \frac{4\pi^2 \rho A}{1.875^4 I} \cdot \text{slope}. \tag{7}$$

Similarly to the first scenario, uncertainties were estimated using Monte Carlo propagation.

2 Fourier analysis

In mathematics, a Fourier series is a method through which a periodic function can be defined as a linear combination of orthogonal trigonometric basis functions. These functions can be decomposed into sine and cosine functions, or equivalently expressed in complex form. Fourier analysis can be used to represent functions defined in time domain or in a spatial-domain by using frequency-domain representations. This forms the foundation of many techniques in physics and engineering, including signal processing, vibration analysis, heat transfer, and quantum mechanics. Fourier analysis also provides a systematic and objective way to extract dominant frequencies from recorded data and thus infer the nature of the oscillatory system being measured.

Fourier analysis was introduced by Joseph Fourier in the early 1800's when he was working on heat flow modeling. Fourier's original theory was controversial and considered mathematically unacceptable at the time; it violated the mathematical standards for defining terms and using conceptual definitions to generate results for situations involving exponentially increasing numbers of mathematical terms.

Later, other mathematicians such as Dirichlet, Riemann, and Lebesgue were able to sort out some of these mathematical issues and develop a more formalized way of explaining Fourier's ideas about how to represent discontinuous functions. For example, Dirichlet developed the conditions required for a Fourier series to converge based on the approximation of a periodic function using an infinite number of terms; Leibniz expanded the theory of integration in later years to allow for use in a broader range of types of function and on the intervals on which those types of function can be evaluated. In the twentieth century, generalizations of Fourier analysis, such as the Fourier transform, were developed to study non-periodic functions and to continue the development of continuous spectra.

Today, the development of Fourier analysis opens a new era of analysis ranging from engineering to medical science.

2.1 Mathematical Formulation of the Fourier Series

If we define a periodic function $f(t)$ with period T, such that

$$f(t+T) = f(t). \tag{8}$$

Then, the Fourier series of $f(t)$ could be written in trigonometric form as

$$f(t) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left[a_n \cos\left(\frac{2\pi n}{T} t\right) + b_n \sin\left(\frac{2\pi n}{T} t\right) \right], \tag{9}$$

where a_0, a_n, and b_n are the Fourier coefficients. These coefficients are given by

$$a_0 = \frac{2}{T} \int_{t_0}^{t_0+T} f(t)\, dt, \tag{10}$$

$$a_n = \frac{2}{T} \int_{t_0}^{t_0+T} f(t) \cos\left(\frac{2\pi n}{T} t\right)\, dt, \tag{11}$$

and

$$b_n = \frac{2}{T} \int_{t_0}^{t_0+T} f(t) \sin\left(\frac{2\pi n}{T} t\right)\, dt. \tag{12}$$

These coefficients quantify the amplitudes of the sinusoidal components in a Fourier series. The term with $n = 1$ is called the fundamental frequency, while the terms with $n > 1$ are known as higher harmonics. The angular frequency of the nth harmonic is

$$\omega_n = \frac{2\pi n}{T}. \tag{13}$$

Applying the Fourier coefficients in Eq. (**??**) and using the following formulas, we can verify these coefficients. For integers m and n, the following relations :

$$\int_{t_0}^{t_0+T} \cos\left(\frac{2\pi n}{T}t\right) \cos\left(\frac{2\pi m}{T}t\right) dt = \begin{cases} 0, & n \neq m, \\ \frac{T}{2}, & n = m > 0, \\ T, & n = m = 0, \end{cases} \tag{14}$$

$$\int_{t_0}^{t_0+T} \sin\left(\frac{2\pi n}{T}t\right) \sin\left(\frac{2\pi m}{T}t\right) dt = \begin{cases} 0, & n \neq m, \\ \frac{T}{2}, & n = m > 0, \\ 0, & n = m = 0, \end{cases} \tag{15}$$

and

$$\int_{t_0}^{t_0+T} \sin\left(\frac{2\pi n}{T}t\right) \cos\left(\frac{2\pi m}{T}t\right) dt = 0. \tag{16}$$

allow the Fourier coefficients to be determined by projection of the function onto each basis function. Dirichlet conditions must be satisfied to construct the Fourier series:

1. The integral of $|f(x)|$ must converge over one period,

$$\int_0^T |f(t)|\, dt < \infty.$$

2. In one period, $f(x)$ must have a finite number of maxima and minima.
3. In one period, $f(x)$ has only a finite number of discontinuities, and each discontinuity is finite

For a periodic function that is piecewise continuous and has a finite number of maxima, minima, and discontinuities over one period, the Fourier series converges to the function value at points of continuity. At a point of discontinuity, the Fourier series converges to the midpoint of the jump:

$$\frac{f(t^+) + f(t^-)}{2}. \tag{17}$$

The Fourier series can also be written in complex form:

$$f(t) = \sum_{n=-\infty}^{\infty} c_n e^{i2\pi nt/T}, \tag{18}$$

where the complex Fourier coefficients are defined by

$$c_n = \frac{1}{T} \int_{t_0}^{t_0+T} f(t) e^{-i2\pi nt/T}\, dt. \tag{19}$$

The trigonometric and complex forms are equivalent. However, in some areas, it is much more common and convenient to use a compact form of Fourier series, due to the complex of problems like signal analysis.

2.2 Fourier Transform

The Fourier transform converts complicate problems into a solvable version. In opposite to the Fourier series, which is applying to a periodic functions, the Fourier transform is meant to extract the imaginary frequencies of a signal in the general case (it could be non-periodic). In other words, it decomposes a function into a continuous spectrum of frequencies. For a function $f(t)$, the Fourier transform is defined as

$$F(\omega) = \int_{-\infty}^{\infty} f(t) e^{-i\omega t}\, dt, \tag{20}$$

and the inverse Fourier transform is

$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) e^{i\omega t}\, d\omega. \tag{21}$$

Alternatively, in terms of frequency f in hertz rather than angular frequency ω, the transform pair may be written as

$$\hat{f}(\nu) = \int_{-\infty}^{\infty} f(t) e^{-i2\pi\nu t}\, dt, \tag{22}$$

$$f(t) = \int_{-\infty}^{\infty} \hat{f}(\nu) e^{i2\pi\nu t}\, d\nu. \tag{23}$$

The frequency domain representation of a signal is obtained by transforming the signal from the time domain into the frequency domain. The magnitude $|F(\omega)|$ describes the strength of the frequency components, and the phase shows their relative timing. This representation is particularly important, as measured time-domain signals often contain multiple oscillatory components, noise, or transient behavior.

2.3 Discrete Fourier Transform and Fast Fourier Transform

In experimental and computational applications, signals are typically sampled at discrete time intervals. For a sequence of N sampled data points $x_0, x_1, \ldots, x_{N-1}$, the discrete Fourier transform (DFT) is defined as

$$X_k = \sum_{n=0}^{N-1} x_n e^{-i2\pi kn/N}, \qquad k = 0, 1, \ldots, N-1. \tag{24}$$

The inverse discrete Fourier transform is given by

$$x_n = \frac{1}{N} \sum_{k=0}^{N-1} X_k e^{i2\pi kn/N}, \qquad n = 0, 1, \ldots, N-1. \tag{25}$$

The DFT converts a finite sequence of sampled data from the time domain to its representation in the discrete frequency domain. The DFT is computed efficiently using the fast Fourier transform (FFT). The FFT is not a different transform but can be considered an algorithm that significantly reduces the computational cost of evaluating the DFT, making Fourier based analysis practical for large data sets.

For a signal sampled with time interval Δt, the sampling frequency is

$$f_s = \frac{1}{\Delta t}, \tag{26}$$

and the corresponding frequency resolution is

$$\Delta f = \frac{f_s}{N} = \frac{1}{N\Delta t}. \tag{27}$$

The highest frequency that can be resolved without aliasing is the Nyquist frequency,

$$f_{\text{Nyquist}} = \frac{f_s}{2}. \tag{28}$$

These relationships are critical in signal analysis because they define the frequency range that can be measured and how accurately spectral peaks can be identified.

2.4 Zero Padding

In discrete Fourier analysis, the spectral resolution of a spectrum can be improved by appending zeros to the end of the signal. This method is called the zero padding method.It should be noted that no additional information will be added or removed from the signal. By extending the signal through this method, the frequency spacing becomes

$$\Delta f_{\mathrm{pad}} = \frac{1}{N_{\mathrm{pad}}\Delta t}, \tag{29}$$

which is smaller than the original spacing by increasing the number of samples to a multiple of two. This produces a denser sampling of the frequency, which leads to a smoother spectral peak. As a result, it can boost the accuracy of peak estimation. In other words, zero padding provides a finer frequency grid, allowing the peak location to be determined more precisely despite the increased width.

In this work, the frequency estimation process is being refined by applying zero padding before the estimation process in order to more accurately identify the dominant resonance frequency of the cantilever system. The density of the number of samples taken will increase, enabling a more accurate identification of the peak location, and when used in conjunction with various types of interpolation/curve-fitting techniques, such as sinc or Gaussian fitting, can improve accuracy.

Figure 1 demonstrates how the zero-padding method effects on the original signal in both time and frequency domain. It can be observed that the zero padding does not alter the original signal within the measurement interval, but extends it by appending zeros. Additionally, in the frequency domain, increasing the padding factor from 2 to 32 refines the sampling of the Fourier spectrum, which is producing a smoother and more densely sampled peak, so, this improves peak localization.

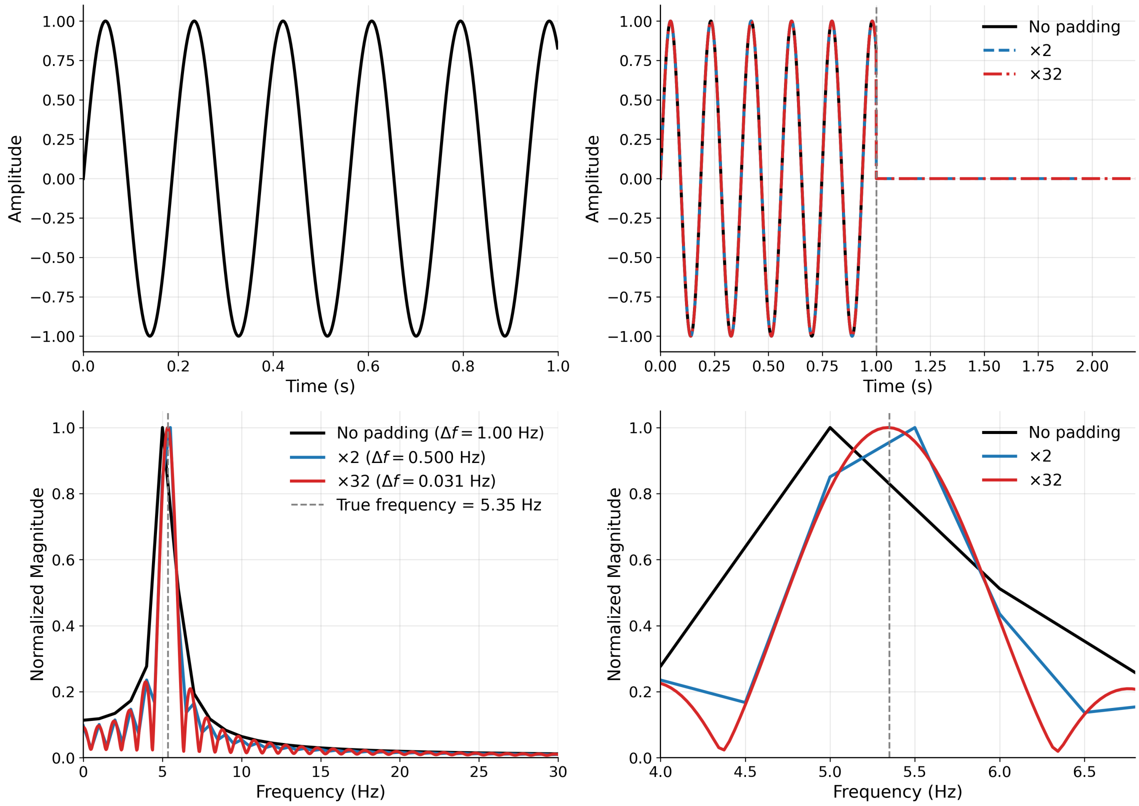

Figure 1: Effect of zero padding on a finite sinusoidal signal. The top panels show the signal before and after adding zeros (right and left panels show the original signal and padded signals, respectively. The bottom panels compare the corresponding Fourier spectra for both entire spectrum(left panel) and zoomed around peak(right panel).

2.5 Apodization (Windowing)

Fourier analysis in practical applications uses a finite measurement of the signal instead of using an infinite or perfectly periodic domain as the analysis domain. By truncating the signal with a finite-duration measurement, a discontinuity is created at the boundaries of the measurement window, which results in energy from the true-frequency component of the signal spreading to neighbouring frequency bins, known as spectral leakage.

As a result, it can be more difficult to clearly identify the dominant frequency components in the Fourier transform of finite-length signals. Even when a windowing function has not been explicitly applied to the signal, this effect is created by the implicit use of the rectangular window when measuring the signal for finite-duration. Therefore, the frequency-domain representation of finite-length signals will typically not show ideal, delta-like peaks but will produce broader spectral features.

To mitigate spectral leakage, a technique known as apodization, or windowing, is applied. This involves multiplying the original signal by a smooth window function that gradually reduces the amplitude of the signal toward zero at the boundaries. For a continuous signal $x(t)$ and window function $w(t)$, the windowed signal is defined as

$$x_w(t) = x(t)\,w(t). \tag{30}$$

In the discrete case, for a sampled signal x_n with $n = 0, 1, \ldots, N-1$, the windowed signal becomes

$$x_n^{(w)} = x_n\,w_n. \tag{31}$$

Several window functions are commonly used, each providing a different balance between spectral leakage suppression and frequency resolution:

- **Rectangular window:**

$$w_n = 1. \tag{32}$$

This corresponds to no apodization and preserves the narrowest spectral peaks, but results in significant spectral leakage due to abrupt truncation.

- **Hann window:**
$$w_n = \frac{1}{2}\left(1 - \cos\left(\frac{2\pi n}{N-1}\right)\right). \tag{33}$$
This window reduces spectral leakage by smoothly tapering the signal, although it broadens the resulting frequency peaks.

- **Hamming window:**
$$w_n = 0.54 - 0.46\cos\left(\frac{2\pi n}{N-1}\right). \tag{34}$$
This window provides improved side-lobe suppression compared to the Hann window, at the expense of slightly reduced frequency resolution.

- **Tukey window:**
The Tukey window combines a flat central region with cosine tapers at the edges, controlled by a parameter $\alpha \in [0, 1]$:
$$w_n = \begin{cases} \frac{1}{2}\left[1 + \cos\left(\pi\left(\frac{2n}{\alpha(N-1)} - 1\right)\right)\right], & 0 \le n < \frac{\alpha(N-1)}{2}, \\ 1, & \frac{\alpha(N-1)}{2} \le n \le (N-1)\left(1 - \frac{\alpha}{2}\right), \\ \frac{1}{2}\left[1 + \cos\left(\pi\left(\frac{2n}{\alpha(N-1)} - \frac{2}{\alpha} + 1\right)\right)\right], & (N-1)\left(1 - \frac{\alpha}{2}\right) < n \le N-1. \end{cases} \tag{35}$$
For $\alpha = 0$, the Tukey window reduces to a rectangular window, while for $\alpha = 1$, it becomes equivalent to a Hann window. Intermediate values allow a tunable balance between leakage suppression and peak broadening.

The consequences of the various window methods, both in the frequency domain and in the time domain, are graphically illustrated in Figures 2 and 3. In Figure 2, the analysis of the time domain shows how various windows affect the amplitude of a signal at the edges of the window, while in Figure 3, the frequency domain shows how windows affect the spectral content.

The analysis of the frequency domain clearly demonstrates the trade-offs associated with using a windowing function. The rectangular window produces the narrowest peak, but has long spectral tails due to high levels of leakage. The Hann and Hamming windows effectively suppress these side lobes but result in much broader peaks and, therefore, a lower frequency resolution. The Tukey window has both characteristics, producing a relatively narrow peak and having lower levels of leakage than the rectangular window.

The Tukey window functions as a good compromise for the finite duration oscillatory signals investigated in this study (signals produced by cantilever beam vibrations). These experimental signals are neither periodic nor stationary; therefore, they have boundaries that create artifacts, which will distort their spectral quality when truncated. By tapering completely on the signal's edges but leaving a flat section in the center, the Tukey window is able to reduce the leakage from the central peak while maintaining the width of the central peak without causing excessive broadening. This will improve the ability to identify fundamental frequencies with accuracy when there is a damping or finite time for observation.

To eliminate ambiguity in comparing the Fourier spectra presented in this section, we normalized the Fourier spectra by their maximum magnitude, so that differences in spectral shape and leakage behavior can be emphasized without the added complexity of absolute amplitude scaling.

Windowing coupled with the zero-padding method significantly increases the accuracy of the peak estimation.

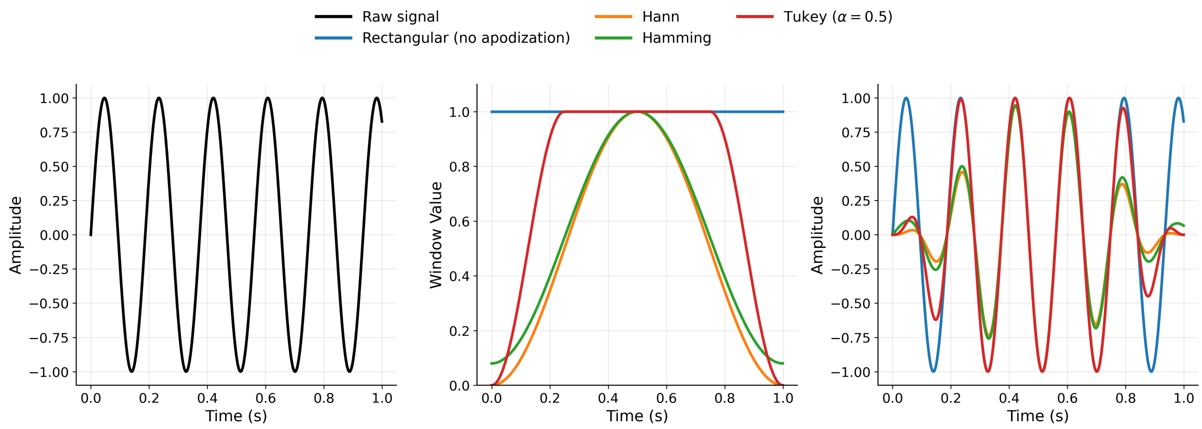

Figure 2: Time-domain comparison of the original signal(left panel), the window functions(middle panel), and the resulting windowed signals(right panel).

3 Uncertainty Analysis

Accurate estimation of Young's modulus requires consideration of uncertainties in experimental measurements, signal processing, and model fitting.

3.1 Geometric Uncertainties

Uncertainty in beam dimensions directly affects Young's modulus estimation. The measured quantities and associated uncertainties used throughout the analysis were:

$$dL = 5 \times 10^{-5} \text{ m}, \tag{36}$$

$$db = 5 \times 10^{-5} \text{ m}, \tag{37}$$

$$dh = 5 \times 10^{-5} \text{ m}, \tag{38}$$

$$dm = 5 \times 10^{-5} \text{ kg}, \tag{39}$$

where L is the beam effective length, b is the beam thickness, h is the beam width, and m is the beam mass.

The second moment of inertia for the rectangular beam cross section was computed as

$$I = \frac{hb^3}{12}, \tag{40}$$

which is sensitivity to thickness due to the cubic dependence on b.

Multiple measurements of beam dimensions were obtained to reduce random measurement errors and provide realistic estimates of dimensional uncertainty.

3.2 Frequency Extraction Uncertainty

Natural frequencies were obtained using Fourier analysis of the measured vibration signals. The time-domain data were detrended, windowed, and transformed into the frequency domain using Fast Fourier Transform (FFT).

To improve frequency resolution beyond the FFT bin spacing, a local sinc-function fit was applied around the dominant spectral peak. This procedure provides sub-bin estimation of resonance frequency and reduces discretization error associated with the FFT.

Repeated measurements were performed for each loeaded case (10 times for each masses). The uncertainty in frequency was estimated from the statistical spread of repeated trials:

$$\sigma_f = \sqrt{\frac{1}{N-1}\sum_{i=1}^{N}(f_i - \bar{f})^2}, \tag{41}$$

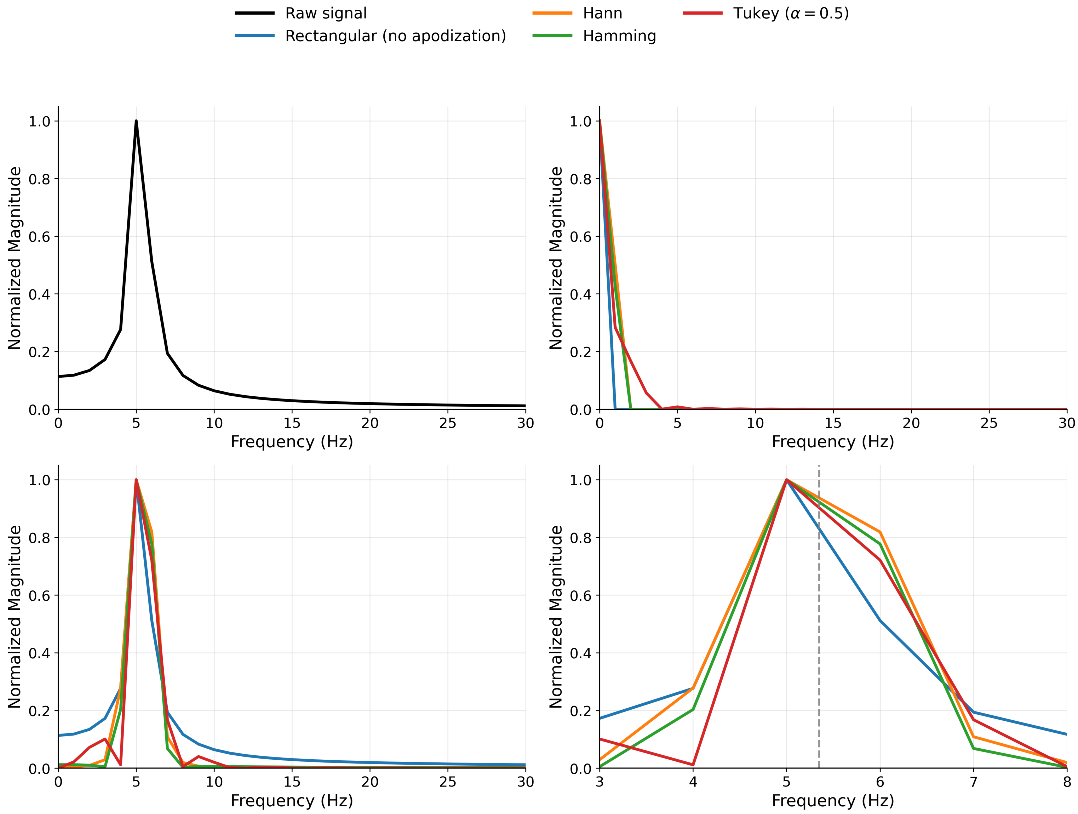

Figure 3: Frequency-domain comparison of the clean signal(top left panel) and the windowed signals (top right panel) after FFT. Bottom left and right panels show the entire The zoomed around peak, respectively.

where N is the number of repeated measurements and $\bar{f}$ is the mean frequency.

3.3 Regression Uncertainty

For the tip-mass method, Young's modulus was obtained from the slope of a linear fit between $\frac{1}{f^2}$ and applied tip mass. For the massless method, the linear fit was performed between f_0^2 and $\frac{1}{L^4}$. Linear regression introduces uncertainty in the estimated slope due to scatter in experimental data.

3.4 Monte Carlo Uncertainty Propagation

Due to their nonlinear dependence upon a range of fitted and geometric parameters, analytical propagation of uncertainty in Young's modulus could be challenging. Therefore, Monte Carlo simulation was employed for uncertainty propagation.

Random samples of geometric parameters and measured quantities were generated according to their uncertainty distributions. For each realization, Young's modulus was recalculated:

$$E_i = f(L_i, b_i, h_i, m_i, f_i), \tag{42}$$

where the subscript i denotes one Monte Carlo realization.

A large number of simulations were performed, producing a probability distribution for Young's modulus. The final estimate was taken as

$$E = \bar{E} \pm \sigma_E, \tag{43}$$

where $\bar{E}$ is the mean of the simulated distribution and σ_E is its standard deviation.

In short, the reported uncertainties in Young's modulus therefore include contributions from:

- beam length measurements,

- beam width and thickness measurements,
- beam mass measurements,
- frequency extraction uncertainty,
- trial-to-trial variability,
- regression uncertainty,
- nonlinear parameter coupling through Monte Carlo propagation.

4 Young modulus

The Young modulus is a criterion through which we can describe the rigidity of solid state material. This criterion is considered to be the mechanical properties of materials that have linear elastic behavior. The Young modulus can be dependent on temperature depending on the material used. This modulus is defined as the relationship between the stress(force per unit area) and the strain (material's relative deformation). The Young modulus, or elastic modulus, is named after a British scientist Thomas Young in the 19th century. However, this concept was developed by Leonhard Euler in 1727. The first experiments using the modern form of the concept of Young's modulus predate Young's work by 25 years, by Giordano Riccati in 1782. The word "modulus" is derived from the Greek word "modus" meaning measure.

4.1 Stress–strain curve

Stress and strain are two well-known concepts in engineering. Through the relationship between stress and strain, we can have a good estimate of the material strength. This relationship can be measured by compression or tension testing on a test sample. In this experiment, the applied force on the loaded sample is continuously increased. By increasing the force, the sample deformation is measured until the sample shows a sign of fracture. Finally, by finding these values, the stress–strain curve can be plotted.

The amount of deformation of a specimen will be determined by its elastic modulus and geometry (length and cross-sectional area). In engineering designs, information on the behavior of materials is required as a result of the effect of their geometry. Therefore, to remove the effect of geometry, the loading is shown as the stress, and the deformation is illustrated as the strain.

Solid materials deform under load. The elastic deformations belonged to the changes in which the material returned to the initial shape after loading.

5 Result and discussion

The objective of this experiment is to determine Young's modulus E by finding the natural frequency of a cantilever beam, which is made of Carbon Fiber Reinforced Polymer (CFRP). To verify our method, we utilized four samples made with four different materials, Aluminum, Stainless steel, Copper, and Brass. In parallel, we use CFRP, which is our target as the fifth sample. We repeated our experiment for the three lengths for each sample. For each lengths we hanged 10 different masses (20g, 40g, 70g, 100g, 120g, 140g, 160g, 180g, and 200g) at the free end and repeated our measurements in ten trials for each mass. Also, we removed the masses and did experiment again without any mass attached to the free end.

Fig. 4 shows the setup of the experiment. It can be seen that the beam is fixed by two screws at one end and is free to vibrate at the free end. In addition, the whole setup is fixed to the table by two metal clamps. A guitar pickup is used as a coil to generate the magnetic field around the beam. As the beam starts to vibrate (excited by the hammer), according to Faraday's law of induction, Equation (44), a changing magnetic flux induces a voltage, and the magnitude of that voltage is proportional to the rate of change of the flux.

$$\mathcal{E} = -N\frac{d\Phi_B}{dt}, \tag{44}$$

Where $\mathcal{E}$ is the induced electromotive force (emf), N is the number of turns in the coil, Φ_B is the magnetic flux through one loop, and t is time. The minus sign indicates the direction of the induced emf according to Lenz's law.

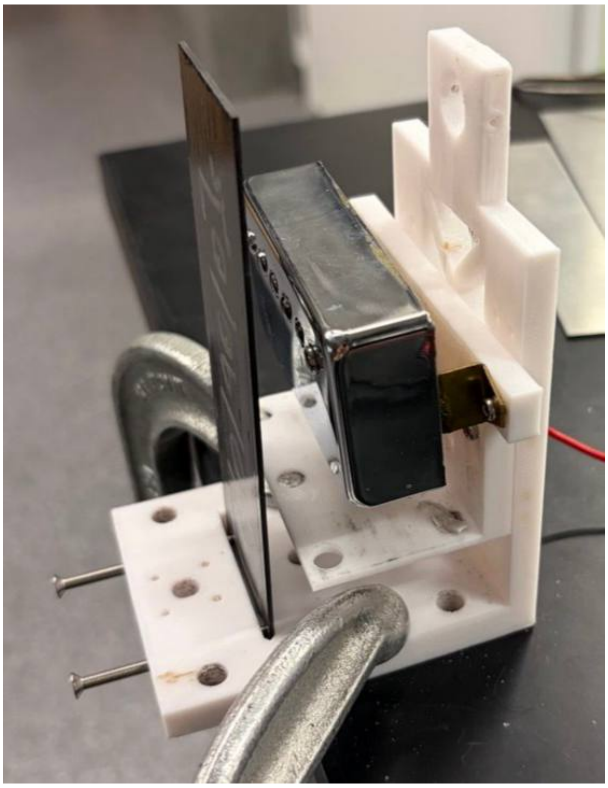

Figure 4: Experiment setup

These changes are recorded by the oscilloscope as a voltage-time signal. The Fast Fourier method is used to convert this signal into the frequency domain. Before getting FFT, zero-padding method and apodization applied to our signal. After finding the spectrum, we fitted a sinc function to the zoomed in area for peak estimation.

Fig. 5 shows the signal in both time and frequency domain for the scenario we attached 120g mass at the free end in trial 1. The top panels show the signal in the time domain before and after applying window. The bottom panels illustrate the signal in the frequency domain, where the fitted sinc function can be seen in the zoomed in plot in the lower right panel.

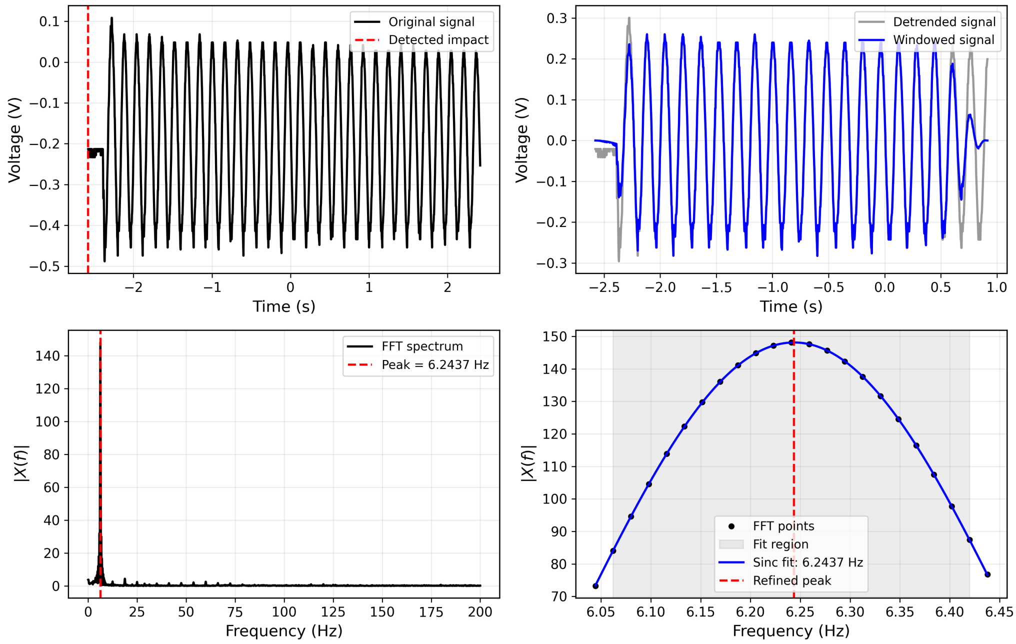

Figure 5: FFT analysis workflow showing (top-left) raw signal with detected impact, (top-right) detrended and windowed signal, (bottom-left) FFT spectrum, and (bottom-right) zoomed peak with sinc-function fit.

After finding the natural frequency, the Young modulus can be found. We use two different scenarios to measure the Young modulus. In the first scenario, we load the free end by hanging masses. In the second scenario, we leave the free end without mass.

For a cantilever beam with a tip mass m_{tip}, the fundamental frequency is given by

$$f = \frac{1}{2\pi}\sqrt{\frac{3EI}{(m_{\text{tip}} + 0.235.m_{\text{beam}})L^3}}, \tag{45}$$

which can be rearranged into a linear form:

$$\frac{1}{f^2} = \frac{4\pi^2 L^3}{3EI}\left(m_{\text{tip}} + 0.235.m_{\text{beam}}\right). \tag{46}$$

Thus, plotting $1/f^2$ versus m_{tip} yields a straight line, where the slope is inversely proportional to E. For a cantilever beam without a tip mass, the fundamental frequency is

$$f_0 = \frac{1.875^2}{2\pi L^2}\sqrt{\frac{EI}{\rho A}}, \tag{47}$$

which leads to

$$f_0^2 = \left(\frac{1.875^4}{4\pi^2}\right)\frac{EI}{\rho A}\frac{1}{L^4}. \tag{48}$$

Therefore, plotting f_0^2 versus $1/L^4$ yields a linear relation, from which E can be extracted.

By comparing both approaches, we assess the consistency of the results and the agreement with theoretical expectations. Measurements were repeated for multiple tip masses and three different beam lengths. Each configuration was measured multiple times to estimate statistical uncertainty.

For each beam length, the quantity $1/f^2$ was plotted against the tip mass. A weighted linear fit was performed to extract the slope. Young's modulus was computed using the relation

$$E = \frac{4\pi^2 L^3}{3I \cdot \text{slope}}. \tag{49}$$

Uncertainties were propagated using a Monte Carlo method, taking into account measurement errors in length, geometry, and fitted parameters. The squared fundamental frequency f_0^2 was plotted as a function of $1/L^4$. A linear fit constrained through the origin was applied. Young's modulus was calculated using

$$E = \frac{4\pi^2 \rho A}{1.875^4 I} \cdot \text{slope}. \tag{50}$$

Similarly to the first scenario, uncertainties were estimated using Monte Carlo propagation. Fig. 6 shows the Young's modulus estimation for aluminum for three different lengths. The dashed-lines represent the values coming from theoretical relations and their corresponding uncertainties are highlighted. The top panel illustrates scenario one where $1/f^2$ is plotted over different hanging mass, where a clear linear relationship between $1/f^2$ and the applied tip mass is observed. Also, a weighted linear line fitted to the data points that allows us to have estimation of Young modulus through its slope. The bottom panel shows the squared natural frequency f^2 is plotted as a function of $1/L^4$, which is the second scenario. the data follow a linear trend consistent with Euler–Bernoulli beam theory:

$$f_0^2 \propto \frac{E}{\rho} \frac{1}{L^4}. \tag{51}$$

The agreement between the slope of the experimental fit and theoretical expectations confirms the validity of the approach. Importantly, the modulus extracted using this method is consistent with the tip-mass results, providing a strong internal cross-check.

Fig. 7 illustrates the comparison across all tested materials. It can be seen that the extracted Young modulus for aluminum, copper, brass, and stainless steel are consistent with expected theoretical ranges within uncertainty band. Both methods provided comparable results. The agreement between these two methods indicates that the test set-up and the analysis pipeline are reliable for testing materials with unknown elastic modulus.

5.1 CFRP: Target Material and Transition to Cryogenic Study

Fig. 8 demonstrates the results for CFRP(with 0/0/90/0 orientation) as our target material, which we assume that the theoretical Young modulus is unknown. At room temperature, the CFRP measurements show a linear behavior in both tip-mass and massless method across multiple beam lengths. Calculating the elastic modulus for CFRP is challenging due to anisotropy of the composite structure, sensitivity to fiber orientation and layup, and deviations from ideal Euler–Bernoulli beam assumptions. Despite these complexities, the agreement between the two independent methods confirms that the measurement pipeline remains applicable to CFRP.

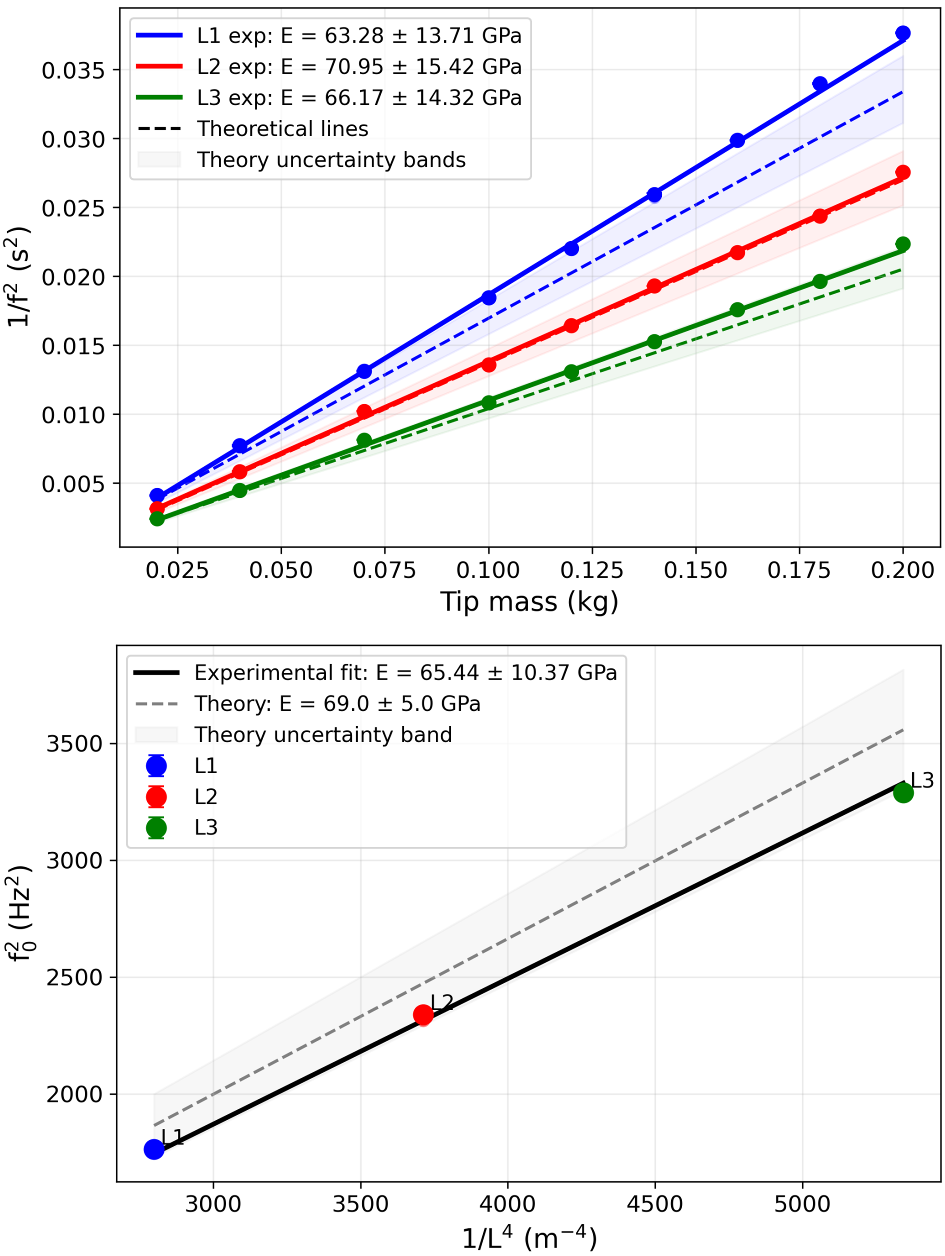

Figure 6: Aluminum results at room temperature: (top) tip-mass method showing linear trends for different lengths, (bottom) massless method showing f_0^2 versus $1/L^4$ with linear fit.

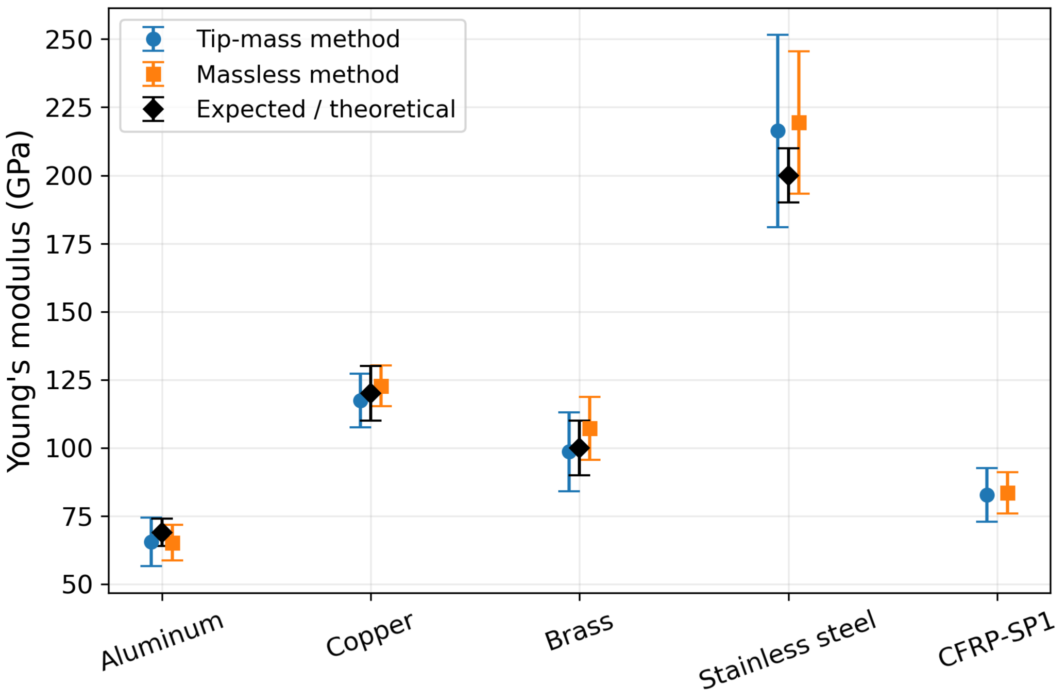

Figure 7: Comparison of Young's modulus across different materials obtained using tip-mass and massless methods. Theoretical values are shown for reference.

6 conclusion

In this work, we built a setup to measure the Young modulus of CFRP. To validate our result, we use different samples with different material in two different methods. Our results show a successful agreements between our experimental and theoretical values in two mentioned methods. The successful validation at room temperature demonstrates that the FFT-based frequency extraction method provides sufficient precision. So, based in the results, the methodology is transferable to composite materials for CFRP and for repeating experiment in cryogenic environment as our next step.

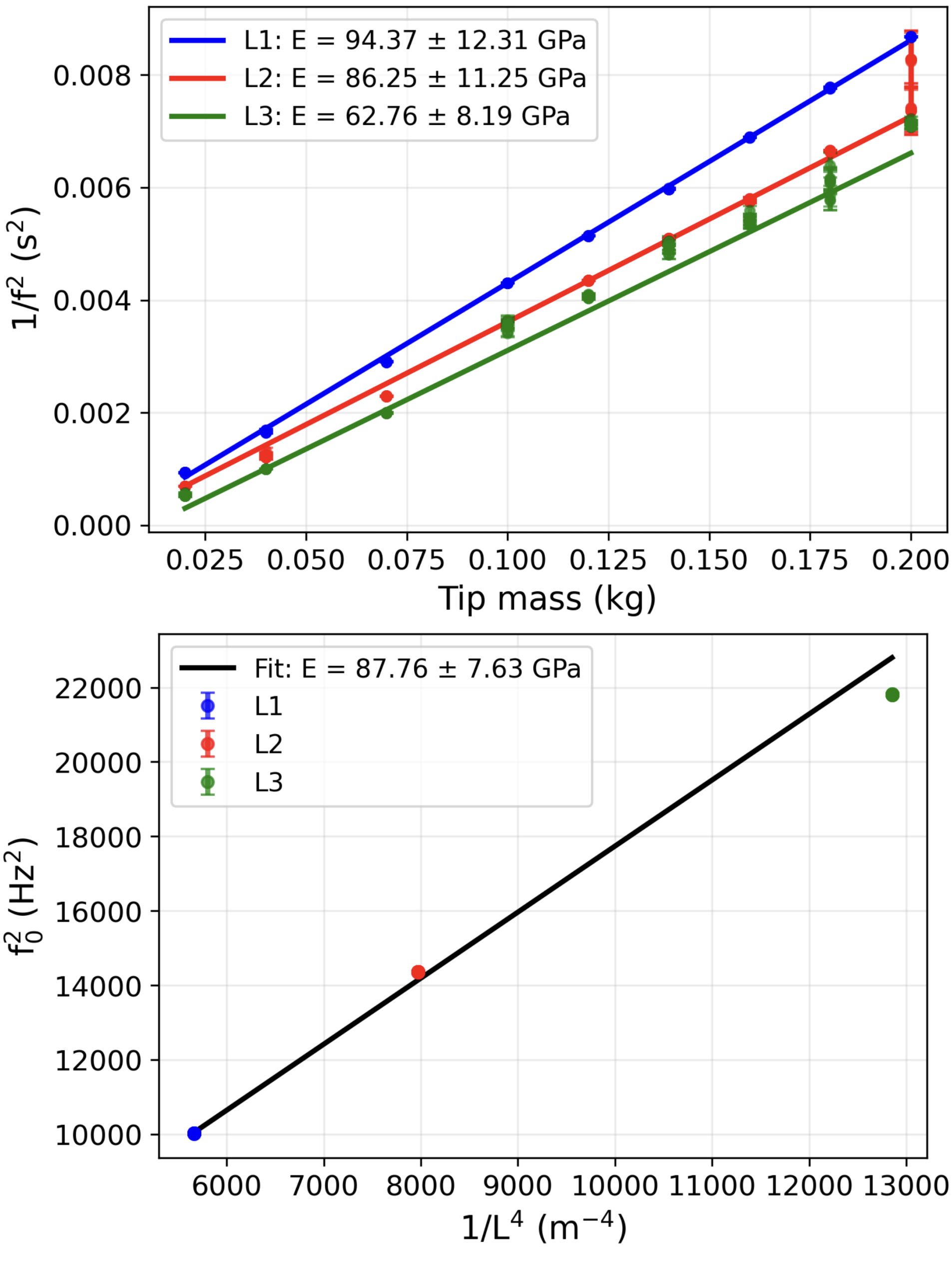

Figure 8: CFRP results at room temperature: (top) tip-mass method showing linear trends for different lengths, (bottom) massless method showing f_0^2 versus $1/L^4$ with linear fit.